

Operating Systems

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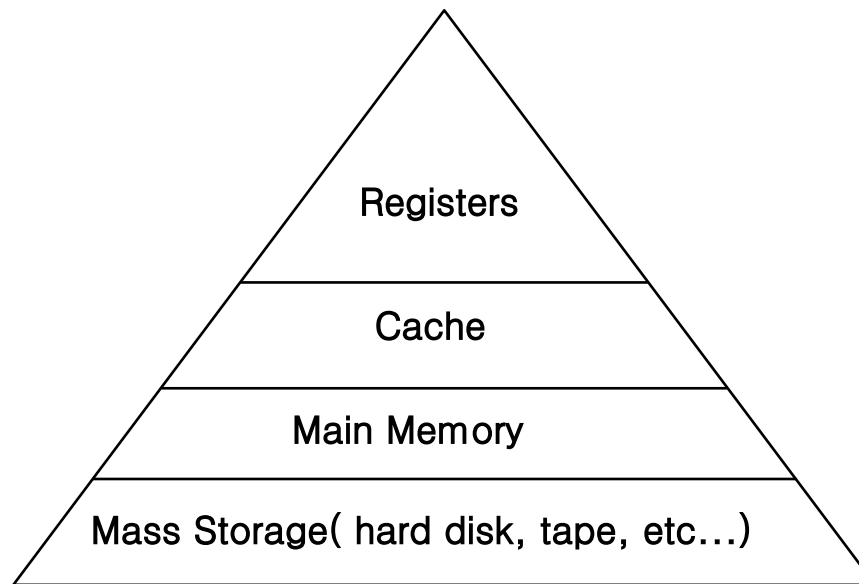
KAIST



21. Swapping: Mechanisms

Beyond Physical Memory: Mechanisms

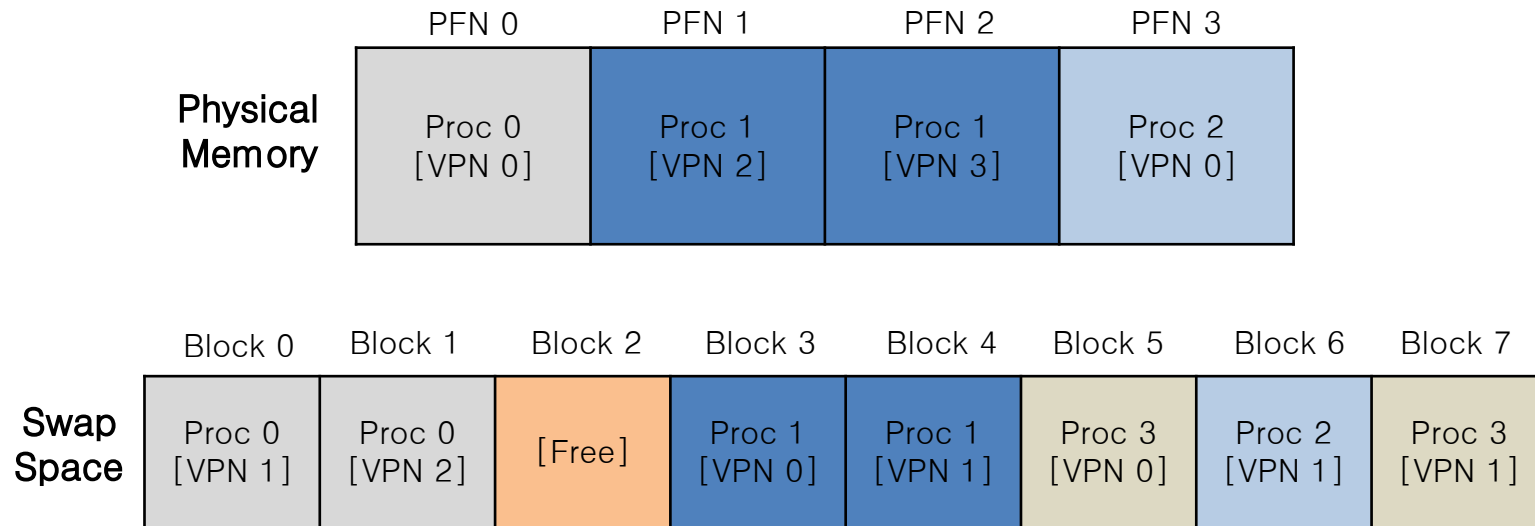
- Use part of disk as memory.
 - ◆ OS need a place to stash away portions of address space that currently aren't in great demand.
 - ◆ In modern systems, this role is usually served by a **hard disk drive**.



Memory Hierarchy in modern system

Swap Space

- Reserve some space on the disk for moving pages back and forth.
- OS needs to remember the swap space, in **page-sized unit**.



Physical Memory and Swap Space

Present Bit

- Add some machinery higher up in the system in order to support swapping the pages to and from the disk.
 - ◆ When the hardware looks in the PTE, it may find that the page is not present in physical memory.

Value	Meaning
1	page is present in physical memory
0	The page is not in memory but rather on disk.

□ Page fault

- ◆ Accessing page that is **not in physical memory**.
- ◆ If a page is not present and has been swapped disk, the OS needs to swap the page back into memory in order to service the page fault.

□ Page replacement

- ◆ The OS likes to page out pages to make room for the new pages the OS is about to bring in.
- ◆ The process of picking a page to kick out, or replace is known as **page-replacement** policy.

When to perform replacement

□ Lazy approach

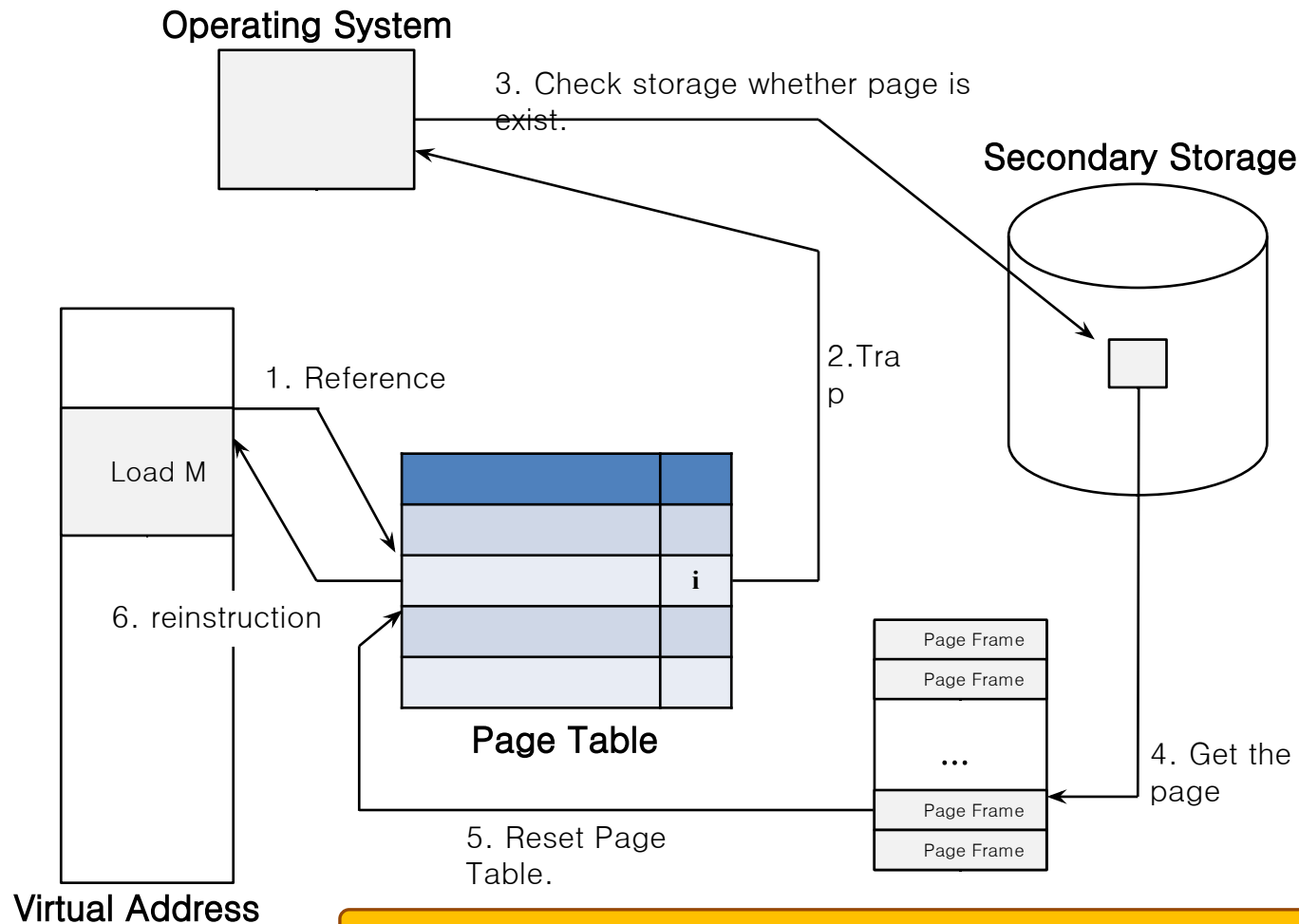
- ◆ OS waits until memory is entirely full, and only then replaces a page to make room for some other page.
- ◆ This is unrealistic.

□ Swap Daemon, Page Daemon

- ◆ There are fewer than **LW (low watermark) pages** available, a background thread that is responsible for freeing memory runs.
- ◆ The thread evicts pages until there are **HW(high watermark) pages** available.

Page Fault Control Flow

- PTE used for data such as the PFN of the page for a disk address.



When the OS receives a page fault, it looks in the PTE and issues the request to disk.

Page Fault Control Flow – Hardware

```
1:  VPN = (VirtualAddress & VPN_MASK) >> SHIFT
2:  (Success, TlbEntry) = TLB_Lookup(VPN)
3:  if (Success == True) // TLB Hit
4:      if (CanAccess(TlbEntry.ProtectBits) == True)
5:          Offset = VirtualAddress & OFFSET_MASK
6:          PhysAddr = (TlbEntry.PFN << SHIFT) | Offset
7:          Register = AccessMemory(PhysAddr)
8:      else RaiseException( PROTECTION_FAULT )
9:  else // TLB Miss
10:      PTEAddr = PTBR + (VPN * sizeof(PTE))
11:      PTE = AccessMemory(PTEAddr)
12:      if (PTE.Valid == False)
13:          RaiseException( SEGMENTATION_FAULT )
14:      else
15:          if (CanAccess(PTE.ProtectBits) == False)
16:              RaiseException( PROTECTION_FAULT )
17:          else if (PTE.Present == True)
18:              // assuming hardware-managed TLB
19:              TLB_Insert(VPN, PTE.PFN, PTE.ProtectBits)
20:              RetryInstruction()
21:      else if (PTE.Present == False)
```

RaiseException(**PAGE_FAULT**)

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Page Fault Control Flow – Software

```
1:   PFN = FindFreePhysicalPage()
2:   if (PFN == -1) // no free page found
3:       PFN = EvictPage() // run replacement algorithm
4:   DiskRead(PTE.DiskAddr, pfn) // sleep (waiting for I/O)
5:   PTE.present = True // update page table with present
6:   PTE.PFN = PFN // bit and translation (PFN)
7:   RetryInstruction() // retry instruction
```

- ◆ The OS must find a physical frame for the soon-be-faulted-in page to reside within.
- ◆ If there is no such page, waiting for the replacement algorithm to run and kick some pages out of memory.

Summary

- ▣ Swapping: making the part of disk as memory
- ▣ Present bit required